Smart Virtual Queue Management System for University Enrolment

Digital ticketing · real-time tracking · predicted wait times to eliminate physical queues

[BHOWAN KHAWAS] · BSc Computer Science · Supervisor: [Dr. Sameena Naaz] · University of Roehampton

Problem Statement

At peak enrolment, University of Roehampton students face long physical queues for ID cards and administrative tasks, with no reliable estimate of waiting. The new physical token system still requires students to stay nearby, causing overcrowding, missed turns, and inefficient staff deployment. This project replaces the physical queue with a digital alternative.

Introduction

Enrolment is a once-per-term bottleneck. Students currently queue physically or collect paper tokens, with no visibility of expected wait — causing stress, lost study time, and operational strain on staff.

This project builds a Smart Virtual Queue Management System: a mobile-friendly web app (PWA) through which students take a digital ticket or book a time slot, track their live position and predicted wait (modelled as an M/M/c queue), and are notified when their turn approaches.

Staff manage the queue via a dashboard, and a public screen shows currently-called numbers. Users are enrolment-office staff and all students enrolling or collecting ID cards. Success is measured against real baseline data and a queueing-theory model.

Aims & Objectives

Aims

1. Design and build a virtual queue system that lets students join enrolment queues remotely and receive live updates.

2. Predict individual wait times using an M/M/c queueing model and measured service data.

3. Evaluate whether virtual queuing + slot booking reduces mean and maximum wait times vs. the current physical token system.

Objectives

1. Measure baseline arrival/service rates and wait times at the enrolment office.

2. Model the enrolment hall as an M/M/c queue; derive wait-time predictions.

3. Build student PWA: digital ticket, live position, predicted wait, notifications.

4. Build staff dashboard: call next, mark served / no-show, manage counters.

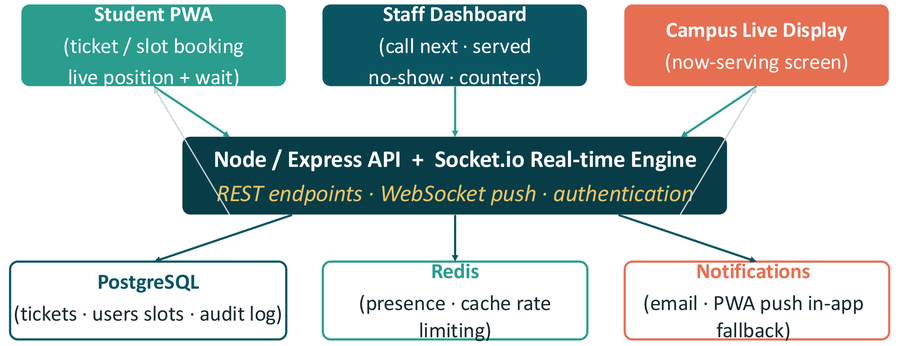
5. Implement a campus live display of currently-called ticket numbers.

6. Build a discrete-event simulation (Python SimPy) comparing walk-in vs virtual vs slot-booking.

7. Evaluate via simulation metrics and a 10–20 participant user study.

8. Deploy prototype with tests, CI/CD, and full documentation.

Proposed Solution — System Architecture



How it works — student journey

1. Join — Take digital ticket or book a slot

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2. Track — Live position + predicted wait

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3. Notify — Alert when turn approaches

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4. Serve — Staff calls next; ticket completed

Technology stack

React PWA · Node / Express · Socket.io · PostgreSQL · Redis · Python SimPy (sim)

Technical core — M/M/c queueing model (the predictive engine)

The enrolment hall is modelled as an **M/M/c queue**: Poisson arrivals (rate λ), exponential service times (rate μ per counter), c parallel counters. Stability requires $\lambda < c\mu$.

Predicted wait is derived via the **Erlang C** formula and **Little's Law ($W = L/\lambda$)**, then calibrated against measured baseline data and validated by a discrete-event simulation.

This gives students a live, data-driven estimated wait — not just a queue position.

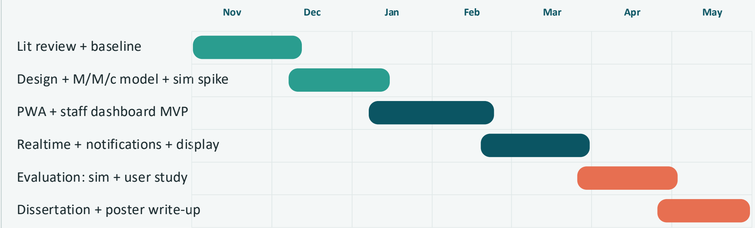
Evaluation — how success is measured

- **Baseline measurement:** Observe real queue at peak: arrival rate, service times, mean/max wait (empirical ground truth).
- **Discrete-event simulation:** SimPy model compares walk-in vs virtual vs slot-booking — quantified wait-time reduction graphs.
- **User study + performance:** 10–20 students (SUS score + feedback); Socket.io latency under simulated concurrent load.

Expected Outcomes & Impact

- **No physical standing queues:** Students join remotely and wait anywhere on campus; overcrowding of enrolment halls eliminated.
- **Data-driven staff allocation:** Peak-time analytics let managers open/close counters and deploy staff efficiently.
- **Transparent, predictable waits:** Live position + M/M/c predicted wait removes uncertainty and reduces student stress.
- **Reusable platform:** Architecture extends to other student services (library, finance, IT helpdesk).
- **Success criteria:** p95 real-time update < 2 s · zero missed notifications · SUS > 70 · simulation shows reduced max wait vs baseline.

Project Plan (Gantt + key dates)



Key dates

- Mid-Nov: baseline measurement complete
- Mid-Feb: working prototype demo
- Mid-Mar: user study conducted
- Early-Apr: dissertation draft
- May: final submission & viva (per university calendar)

Legal, Ethical & Professional

Legal: Student data (names, IDs, enrolment status) handled under UK GDPR / Data Protection Act 2018 — data minimisation, encryption at rest, role-based access, deletion after enrolment.

Ethical: User study requires informed consent, anonymised analysis, right to withdraw; university ethics approval sought.

Professional: Good SE practice — version control, automated tests, CI/CD, documentation; WCAG accessibility; reliability via at-least-once notifications, idempotent tickets, and in-app + email fallback.

Use of AI

AI tools (e.g., GitHub Copilot, ChatGPT) may assist with code scaffolding, documentation drafting, and literature-search summaries. All AI output is reviewed, tested, and rewritten where needed — nothing is submitted without verification. Wait-time predictions are validated against measured data to mitigate inaccuracy, and models are checked for bias across student groups. All AI use is logged and disclosed per university policy.

Key References (IEEE format)

[1] S. Deoraj and M. S. Essop, "Investigating a Virtual Queueing System for Durban University of Technology: A Comprehensive Review Approach to Improve Efficiency," in 2024 Conf. Information Communications Technology and Society (ICTAS), 2024, doi: 10.1109/ICTAS59620.2024.10507144.

[2] T. R. Jadhav, K. B. Adakure, R. A. Thakur, A. M. Thorat, and A. Bhabad, "AI-Based Web Queue Management System for Smart Campus Administration," J. Advance and Future Research, 2026.

[3] J. D. C. Little, "A Proof for the Queuing Formula: L = λW," Operations Research, vol. 9, no. 3, pp. 383–387, 1961.

[4] D. Gross, J. F. Shortle, J. M. Thompson, and C. M. Harris, Fundamentals of Queueing Theory, 4th ed. Hoboken, NJ, USA: Wiley, 2008.